CAPSTONE PROJECT - EMPLOYEE ATTRITION

Group Name:	***9th - 2***
Project Title:	***Employee Attrition***
Date:	***19/02/2026***

Abstract

Employee attrition refers to the replacement of staff over a given period of time and represents a significant loss for companies. In this context, the current project aims to analyze an IBM HR analytics dataset available on Kaggle using machine learning techniques to predict employee attrition. It was observed that the dataset does not contain missing values. Through Exploratory Data Analysis (EDA), it was found that employees in HR and Marketing have higher salaries, that job role is a very important attribute, and that higher attrition was reported among single employees as well as among those who work overtime. Since there were far fewer employees who left the company compared to those who stayed, it was necessary to balance the dataset. Subsequently, five different models were tested (Simple Logistic Regression, KNN, Decision Trees, Random Forest, and SVM), and a grid search was performed. After analyzing accuracy, precision, recall, and F1-score, it was concluded that the best-performing model was simple logistic regression with an average precision of 0,622 after feature engineering.

1. Project Overview

1.1 Problem Statement:

Our project analyzes the topic of employee attrition. Firstly, it's important to delve into what it is, the consequences it poses for companies and corporations and the necessity to further examine this problem. Employee attrition refers to the reduction of the workforce of a business when employees leave. It can be classified as voluntary or involuntary based on whether it was initiated by the employee or by the company. From a corporations' perspective, it's particularly important to understand what causes voluntary attrition. There are two main categories here: on one hand, the employee may leave the company motivated by unforeseen and uncontrollable circumstances such as retirement or illness; On the other hand, the reasons for leaving may be a consequence of underlying issues within the

organization. The rates of employee attrition may also correlate with specific groups of employees. For a company it's very important to understand and address these situations.

It's no secret that high rates of employee attrition are very costly for corporations for several reasons. First, it can result in the loss of highly skilled employees to competitors; It also takes a toll on company productivity, as it's harder to complete the workload while understaffed. It can also negatively affect employee morale as well as customer retention in service-oriented industries.

Lastly, employee attrition represents a great financial loss for the company. There is a great amount of money that goes into recruiting, selecting and training a new employee, as well as overtime from other employees during the process.

1.2 Objectives:

Being able to predict whether an employee is likely to leave can be extremely valuable for corporations, as it allows them to better prepare for such situations. For example, HR departments can identify potential vacancies in advance and begin the process of recruiting and selecting new employees in a timely manner. Additionally, understanding the patterns behind employee attrition allows companies to take preventative measures to ensure retention, thereby saving time and resources.

That being said, the goal of our project is creating a model that predicts employee attrition. For this effect we'll use the following machine learning models: Logistic Regression, K-Nearest Neighbors Classifier (KNN), Decision Tree Classifier, Random Forests and Support Vector Classifier (SVM).

1.3 Literature Review

Employee turnover in a company refers to the replacement of staff over a given period of time, which occurs due to voluntary or involuntary departures. It is worth noting that voluntary resignations often represent the loss of highly skilled employees. In addition, high turnover also requires efforts from Human Resources to find new employees to fill the vacant positions, leading to financial losses and negative impacts on the company's image. According to research by the Society for Human Resource Management (SHRM), it costs a company, on average, 6 to 9 months of an employee's salary to replace them (Enrich, 2024). In this context, predictive models of employee attrition can be highly valuable, as they enable the implementation of strategies to address the situation and prevent the loss of qualified personnel.

A key issue in machine learning is the treatment of imbalanced datasets, in which one class (the majority class) contains significantly more data than the other (the minority class),

often leading to poor predictions for the minority class. In Fang and Zhang (2025), an analysis of an imbalanced HR dataset sourced from Kaggle is presented, in which the number of employees who left the company is much smaller than the number of those who stayed. In this study, two methods were used: the Synthetic Minority Over-sampling Technique (SMOTE) and Generative Adversarial Networks (GAN). The SMOTE method identifies the k-nearest neighbors for each minority sample, computes the difference between samples, and multiplies it by a random value between 0 and 1, which is then added to the original data to generate synthetic samples. On the other hand, the GAN method consists of two neural networks: a Generator and a Discriminator. The Generator analyzes the distribution of the minority class to create synthetic samples, while the Discriminator learns to distinguish fake samples from real ones. Once trained, the Generator is used to create synthetic samples in order to balance both classes in the dataset. The paper evaluated eight different machine learning models: Logistic Regression, Naive Bayes Classifier, Decision Tree, Support Vector Machine, Random Forest, XGBoost, Deep Neural Network, and a hybrid model combining Random Forest and Neural Networks. The results showed that Random Forest and XGBoost achieved the highest overall performance, reaching F1-scores of 0.94, and also demonstrated the best Precision-Recall Curve (PR-AUC) results. Furthermore, the GAN balancing technique improved the performance of all models compared to the baseline, except for the Support Vector Machine. When comparing SMOTE and GAN, GAN showed slightly better performance, indicating superior sample generation.

When it comes to attrition analysis, a commonly used dataset is the IBM HR dataset, which was created by data scientists to study employee attrition. In Amzad Basha et al. (2024), different machine learning models were compared using this dataset. Seven models were employed in the analysis: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting (GBM), XGBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). The results indicated that Random Forest, GBM, and KNN achieved high true negative rates, demonstrating good performance in identifying employees who did not leave the company. However, Random Forest and KNN also exhibited high false negative rates, meaning that many employees who left the company were not correctly identified. Consequently, SVM achieved the best overall balance among the evaluated models.

Although the IBM HR dataset is widely used, it can be limiting because the conclusions derived from it may be specific to the context of that particular company. In addition, the dataset is both fictional and imbalanced, which may reduce the reliability of the findings. Furthermore, it is based on the American population, which may restrict generalizability, as employee behavior can vary across different cultural contexts. In Naik and Chakraborti (2025), the writers used duration of an employee tenure instead of attrition as their target variable, as it allows companies to allocate employees to specific projects based on this. As tenure is a continuous function, this becomes a regression problem rather than a classification one. In order to overcome these issues, the researchers used Linkedin, as a way of collecting data of individuals rather than companies. Using this data also allowed them to analyze the impact of new variables, such as Linkedin connections (more related to social

networks). The premise is that understanding employee profiles is a crucial aspect of the hiring process. Although each employee possesses unique characteristics, it is assumed that, for the purpose of predicting employee churn, employees can be classified into distinct categories based on shared patterns. They did this using: K-Means clustering, Spectral Clustering, and Hierarchical Clustering.

To solve the imbalance class problem, there are different methods that can be used in order to avoid biased predictive models. In P et al. (2022b), researchers used the IBM HR Analytics dataset to evaluate three distinct sampling techniques: Random Oversampling, Random Undersampling, and SMOTE (Synthetic Minority Over-sampling Technique). After balancing the data, they applied five machine learning algorithms: Logistic Regression, K-Nearest Neighbor, Decision Tree, Random Forest, and AdaBoost. In terms of model performance, the results demonstrated that the AdaBoost classifier consistently outperformed other models, achieving the highest AUC scores. Among the sampling methods, SMOTE proved to be the most effective, reaching an Average Precision Recall Score of 73.0, followed by Random Oversampling at 67.6, and Random Undersampling at 51.9. The metric Average Precision Recall Score describes a precision recall curve. The AUC of this curve is an appropriate metric for imbalanced datasets, because the classifier will have a low performance if it is misclassifying most of the minority class. Higher AUC values indicates it has both a high precision and a high recall value, therefore having a low false negative and a low false positive rate. Ultimately, the paper concludes that combining SMOTE with the AdaBoost algorithm leads to the best framework for HR departments to predict employee attrition.

Scikit-learn also provides a parameter called *class_weight*, which is used to handle imbalanced datasets. This method assigns a higher weight to the minority class and a lower weight to the majority class. Setting it to *balanced* automatically adjusts the weights to be inversely proportional to the frequency of each class in the dataset.

Undersampling methods for imbalanced datasets consist of reducing the number of samples in the majority class to achieve a more balanced distribution. The Random Undersampling technique removes majority-class samples at random until the classes are balanced. Another undersampling method is Tomek Links, which identifies pairs of samples from different classes (one from the majority class and one from the minority class) that are each other's nearest neighbors. The algorithm can then remove the majority-class sample or both samples in the pair. The Edited Nearest Neighbours (ENN) method evaluates each majority-class instance by examining its *k* nearest neighbors; if most of its neighbors belong to a different class, the instance is removed. Finally, the NearMiss technique applies rules based on nearest-neighbor distances to select majority-class samples that are closest to minority-class samples, effectively removing those that contribute least to class separation (Imbalanced-learn Developers, 2024a).

Oversampling methods, on the other hand, consist of creating new samples for the minority class. The standard Synthetic Minority Oversampling Technique (SMOTE) considers the *k* nearest neighbors of minority-class instances to generate new synthetic samples through interpolation, using a random value between 0 and 1. A method similar to SMOTE is ADASYN, which also generates synthetic samples but allocates more new samples to minority instances that are harder to learn—those surrounded by many majority-class neighbors. Another variant is Borderline-SMOTE, which focuses specifically on "danger" samples—minority instances whose nearest neighbors are mostly from the majority class. For these borderline cases, new synthetic samples are generated using the SMOTE procedure to reinforce the decision boundary (Imbalanced-learn Developers, 2024b).

When it comes to building and training a machine learning model, feature selection plays an important role, as it prioritizes the most relevant features in the dataset, improves model performance, and reduces computational demand. In Shakir (2025c), the authors developed a technique called Fairness-Aware Mutual Information, which enables a feature selection method that explicitly takes fairness into account. The idea behind this approach is to mitigate bias before model training. To maximize fairness, the method uses the Equal Opportunity Difference (EOD) metric, which measures the difference between the true positive rates of a privileged group and an unprivileged group. In this way, the subset of features used in the final model allows for a less biased approach without compromising other metrics, such as accuracy.

2. Dataset

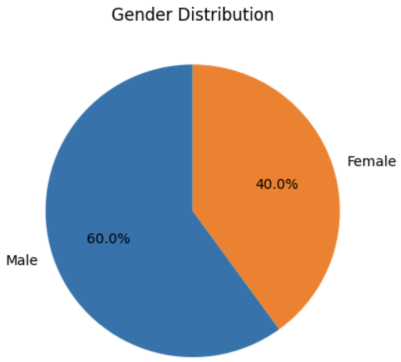
This project uses the IBM HR Analytics dataset from Kaggle, which consists of 1,470 entries (employees) and 34 variables, including information on employee attrition and various factors that may influence employee behavior. It has 26 numerical and 7 categorical columns. The features pertain to a wide variety of factors: employee background characteristics, job information, salary, work experience and conditions, performance and satisfaction. One of the main challenges of this project is that the dataset is imbalanced, with the minority class accounting for approximately 16% of the observations. It's important to understand what type of company is represented on the dataset, in order to get a better understanding of context, data features and patterns and how they relate to this particular business model. The two biggest departments were Research & Development and Sales, and the most common Education Fields amongst the employees were Life Sciences and Medical. This strongly suggests the company is likely in Pharmaceuticals, Biotechnology or other related areas.

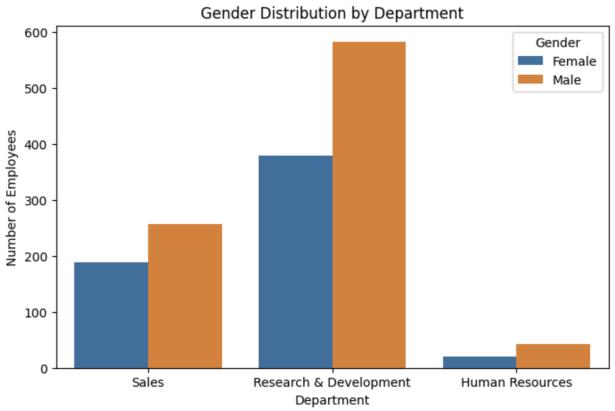
2.1 Exploratory Data Analysis

2.1.1 Initial Analysis

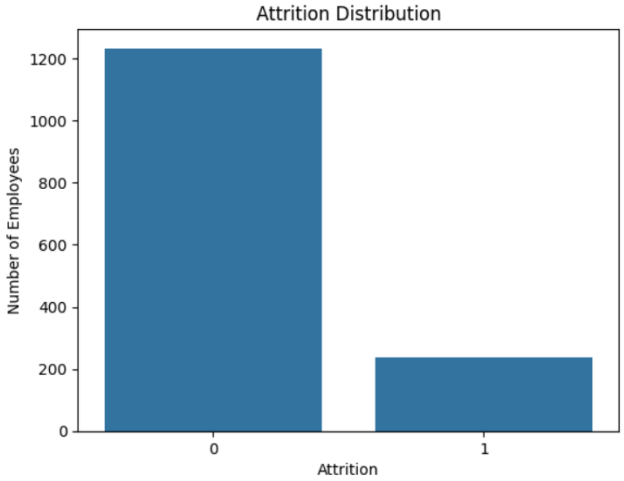
We started by verifying the quality of the dataset by checking the null, missing and duplicated values, determining that there are none. The company is composed of 60% men and 40% women. Women are underrepresented across all departments.

Gender	count
Male	882
Female	588



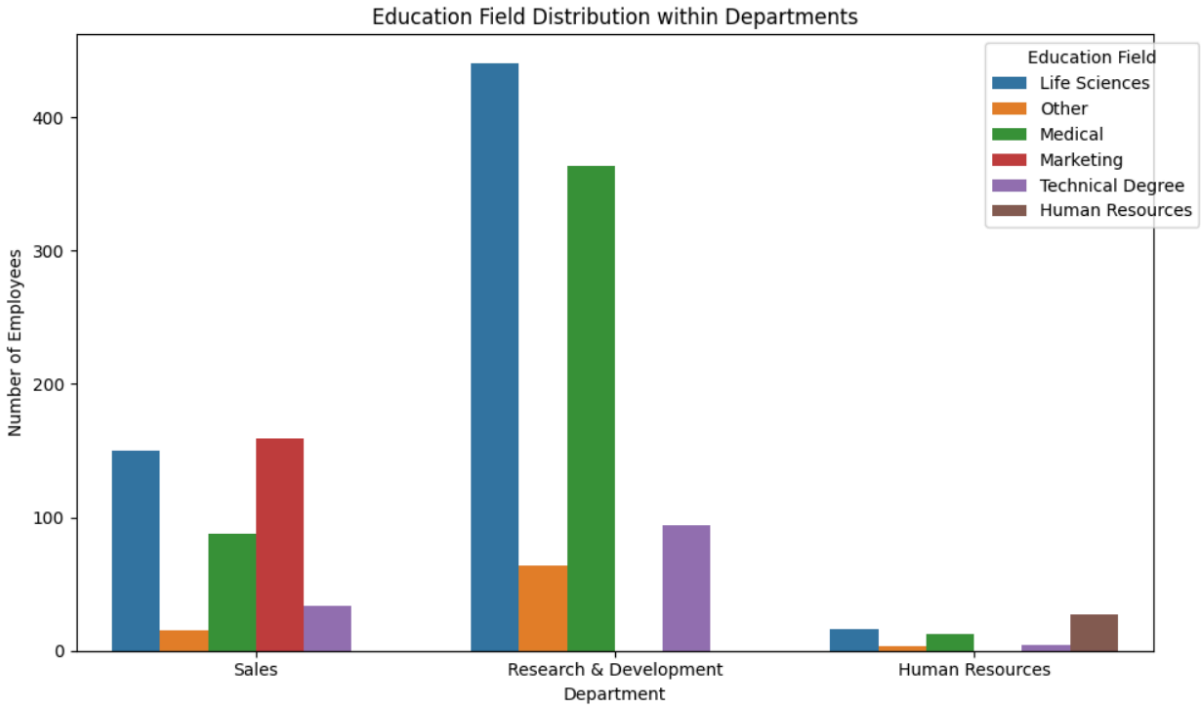


We plotted the attrition distribution in the company to check the frequency of each of the two possible classes, 0 and 1, which correspond to staying at the company and leaving the company, respectively.



The figure above shows that the attrition rate is around 16% so the dataset is very unbalanced.

We can also look at some of the features to figure out what type of company this could be to give us a better understanding of the potential challenges that could be faced. Some features, such as Education Field, Department and Monthly Income usually reveal a lot about a company and the people who work there, so we further analyzed them to search for answers.

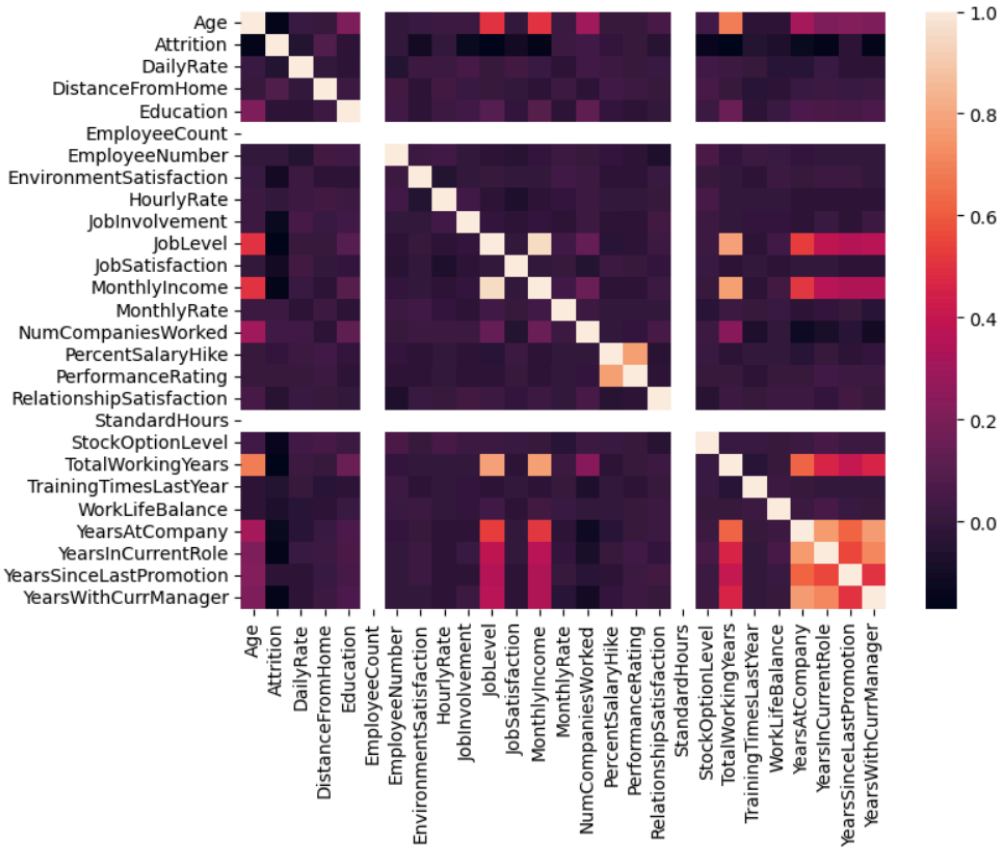


Given that this company has a Sales and a Research & Development department, it should be a product-oriented company. The high number of employees with Life Sciences and Medical Degrees could also be an indicator that they operate in the Pharmaceutical or BioTech industries. These industries are typically very competitive and require employees that are highly educated, so employee retention is critical.

The company has a payroll of almost 9.5M $, calculated by summing all values in the Monthly Income column, suggesting that this is a large company. For organizations of this size, employee attrition can be particularly costly due to recruitment expenses, onboarding time, productivity loss, and potential disruption to operations. Maintaining low attrition is therefore critical for preserving operational efficiency and sustaining competitive advantage.

2.1.2 Correlation Analysis

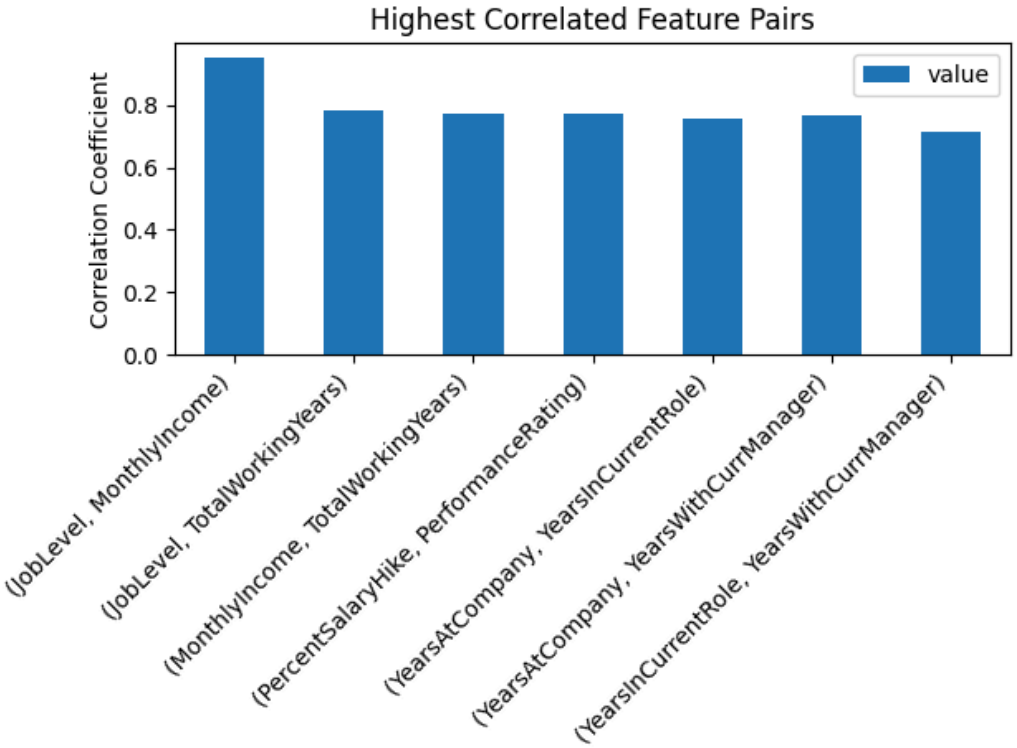
Afterwards we checked the correlation of the features using a heatmap to get a better understanding of any existing patterns or relationships.



Some initial observations from analyzing the heatmap are:

- Age has some correlation with JobLevel, MonthlyIncome and TotalWorkingYears;
- MonthlyIncome and JobLevel have the highest correlation of all, while MonthlyIncome and TotalWorkingYears also have a strong correlation;
- YearsAtCompany, YearsInCurrentRole and YearsWithCurrManager have very strong correlations with each other which might indicate that they could be redundant;

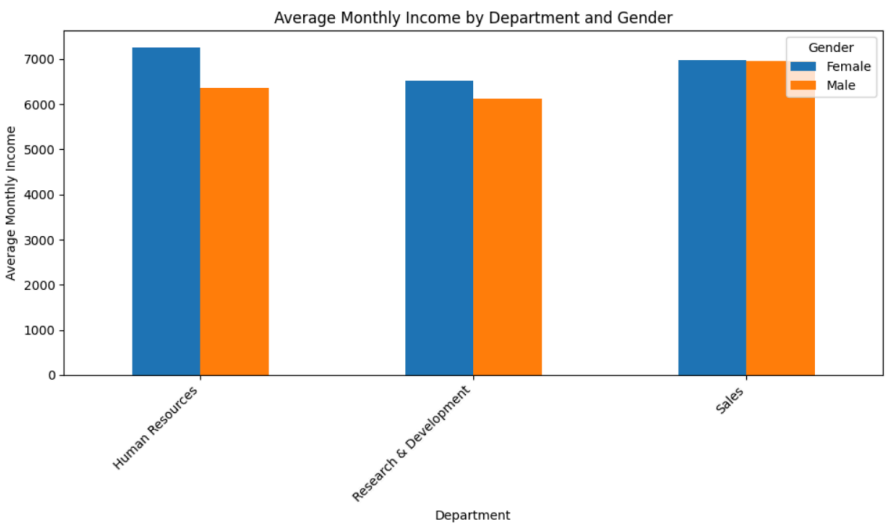
A correlation between 0.7 and 0.95 generally indicates a strong linear relationship, suggesting that the features might contain redundant information. We isolated the feature pairs that meet this criteria.



There are seven pairs within this range that could either lead to data removal, to reduce redundancy, or merging overlapping features.

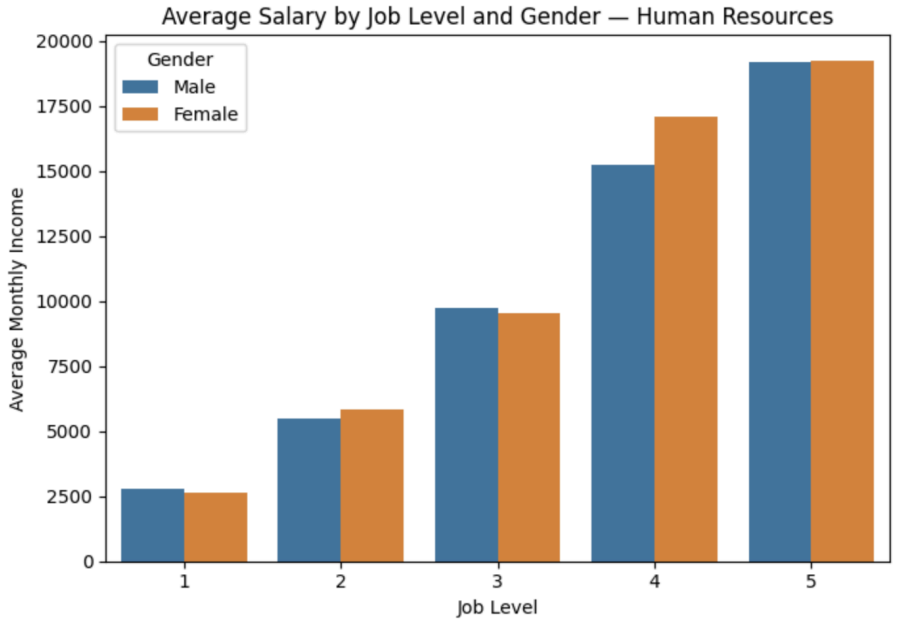
2.1.3 Bias Analysis

After an initial analysis, we have chosen to explore whether any kind of bias exists with respect to a personal factor, like gender, or an organizational factor, like department, starting with plotting the average income by gender in each department.

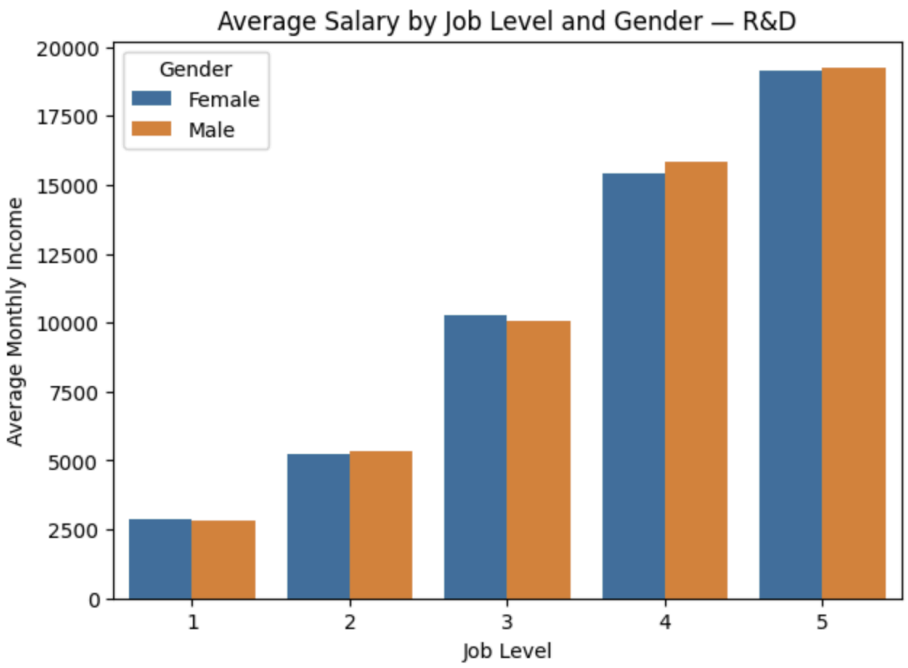


A noticeable difference is observed in the Human Resources department and a slight one in the Research & Development department. In contrast, the Sales department shows nearly identical average salaries, so it is not explored further and we will focus on the other two departments.

We analyzed the internal structure of the Human Resources department by job level and gender, to better understand if there are any significant differences.



At lower job levels, salary differences between men and women are small and inconsistent. However, in higher-paying positions, particularly at Job Level 4, female employees show a higher average salary. This can explain why the average female salary is so much higher.



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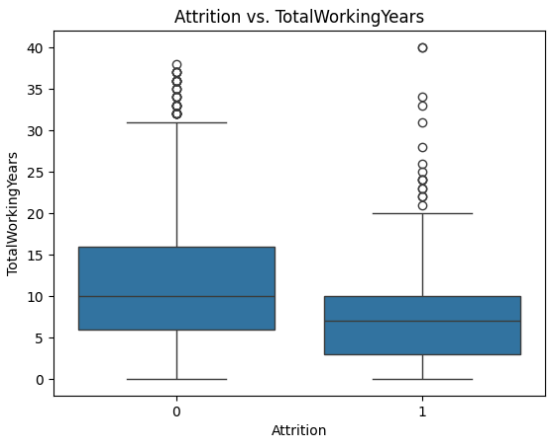
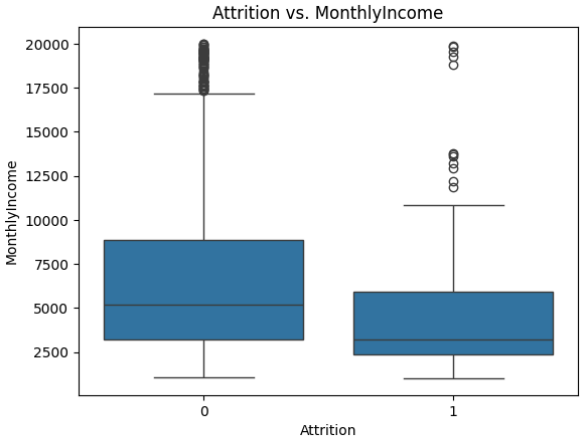
A similar analysis was conducted for the Research & Development department. Salary differences between men and women are small across all job levels, with no consistent pattern favoring either group.

While initial comparisons suggested salary differences between genders in both the Human Resources and Research & Development departments, deeper analysis by job level and role shows that compensation is largely similar between men and women. The observed discrepancies are primarily explained by structural composition effects and the greater influence of higher-paying positions on the overall average, rather than systematic pay inequality.

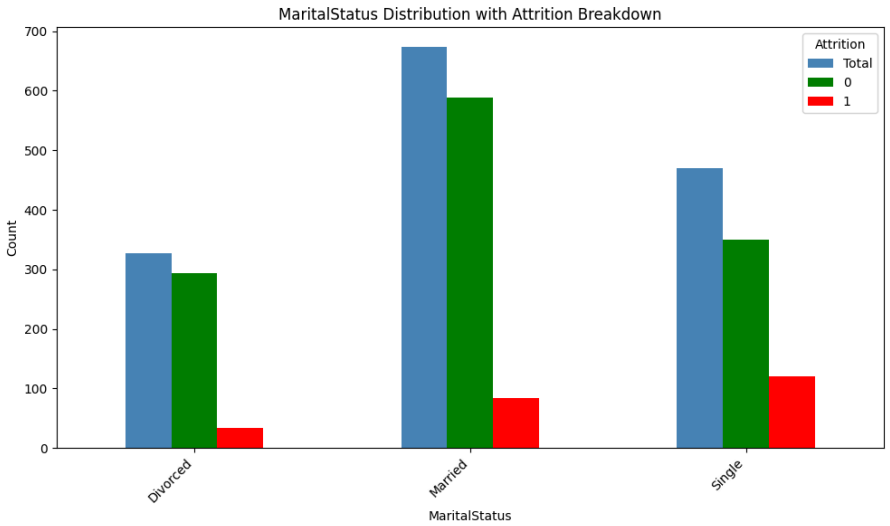
2.1.4 Attrition-Focused Analysis

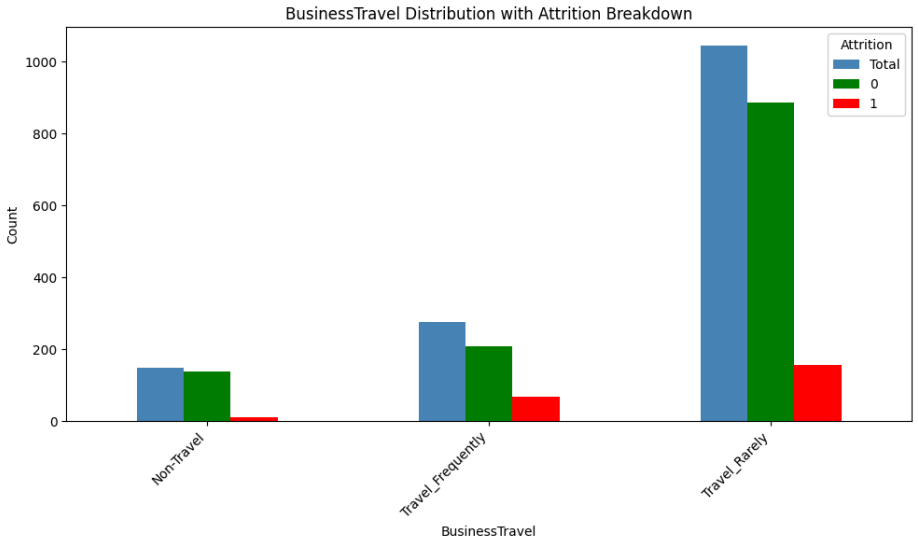
In terms of how certain features affect the attrition rate, we found that some features such as the Total Working Years, Monthly Income, Marital Status, Business Travel and Overtime show strong significance.



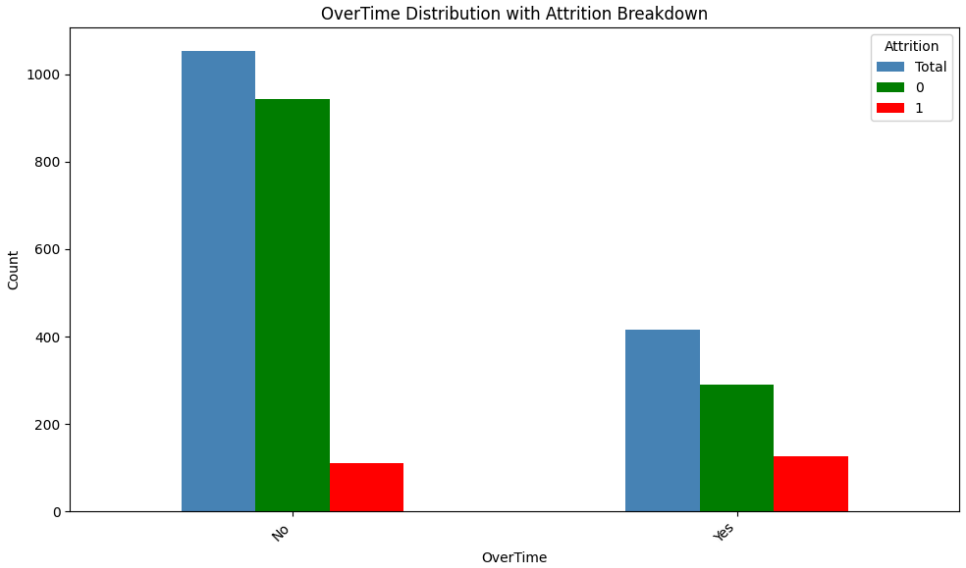


The median work experience of someone who decided to leave the company is more than three and a half years lower than that of the median work experience of someone who decided to stay, which suggests that people with less experience are more likely to quit. Regarding Monthly Income, employees who stayed at the company have higher salaries with the average salary being around 4800$ and 6800$, for people who quit and for those who stayed, respectively. The 2000$ difference is a strong indicator that higher salaries can help retain employees.





The above figures show that single people are more likely to quit than employees who are married or divorced. While only 32% of the employees are single, they make up 51% of everybody who quit. Frequent work travel appears to have a noticeable impact on attrition as well. Even though only 19% of employees travel frequently, they account for 29% of the employees who quit.



Working extra hours seems to be a significant factor for people who quit as well, as shown in the figures above. People who worked overtime make up 28% of all employees and 54% of the people who left the company which indicates a very strong influence.

3. Methodology and Approach

We followed a structured machine learning pipeline to analyze and model employee attrition. The workflow consisted of the following stages: Data analysis, data preprocessing, feature engineering, splitting the data, and finally training and evaluating the models.

3.1 Performance Evaluation Metrics

Accuracy:

Accuracy shows the proportion of correct predictions made by the model out of all predictions. However, for imbalanced datasets it can be misleading. A bad model can predict only the majority class and still get high accuracy, even if it completely ignores the minority class. In our particular case, we can have high accuracy in a model that predicts no one will leave the company, so we chose to focus on other metrics.

Precision:

Precision measures how many of the positive predictions made by the model are actually correct, and is particularly useful when False Positives are costly. For employee attrition, a model that frequently predicts someone will leave when they don’t can pose some problems, such as leading to wasted retention efforts, making this a metric worth analysing.

Recall:

Recall measures how many of the actual positive cases were correctly identified by the model. It’s important to study when False Negatives come at a high cost. Looking again at employee attrition, a model that wrongly predicts an employee will stay when in fact they leave is very ineffective, as it fails to give the company time to take preventative measures. That being said, recall is an important metric to take into consideration.

F1 Score:

F1 Score combines both precision and recall, and a high value indicates it does well in both metrics. It's useful when a balance is needed between both.

ROC-AUC:

The Receiver Operating Characteristic (ROC) curve shows the True Positive Rate vs the False Positive Rate evaluated at different thresholds, representing how well a model can distinguish between two classes. The AUC, which varies between 0 and 1, classifies the model’s ability to differentiate between employees who leave and those who don’t. However, this metric may give overtly optimistic results in imbalanced datasets.

PR-AUC:

The PR curve is more suitable for imbalanced data, as it plots precision and recall at different decision thresholds, exemplifying the tradeoff between the two. It is summarized by a single value by the Average Precision, which represents the area under the curve. The goal is to find a balance between the amount of False Positives and False Negatives. This metric is particularly relevant in our case as an ideal model should avoid incorrectly predicting employees will leave, but also shouldn't fail at identifying when one will. Therefore, we choose this metric to evaluate our model.

3.2 Machine Learning Algorithms

For the Machine Learning algorithms, five different models were tested: Logistic Regression, K-Nearest Neighbors (KNN), Decision Trees, Random Forest, and Support Vector Machine (SVM). These are supervised learning models in which algorithms will learn from a training dataset and based on the features, predict if an employee will leave the company.

Logistic Regression, as the name suggests, is a regression-based classification model. The dependent variable (Attrition) is converted into a numerical value (1 if yes and 0 if no), and the model predicts the probability of attrition occurring. This model has some advantages, as the coefficients can represent feature importance and it has fast training. However, it assumes that the variables are independent and not highly collinear. Therefore, Logistic Regression was chosen as an initial analysis to establish a baseline. The explanation of the parameters used for this model is as follows:

-C: refers to the regularization strength used to prevent overfitting.
-Penalty (L2): a regularization technique that aims to shrink the coefficients toward zero.
-Solver: an optimization algorithm used to minimize the loss function.

K-Nearest Neighbors (KNN) refers to a classification method based on the majority label of the k nearest data points surrounding a specific observation. Thus, it is important to select an appropriate value of k. The advantages of this model are that it is simple and intuitive. However, it does not have a traditional training stage, as it only stores the training data. Therefore, there is no explicit parametric model to analyze, and it requires more memory and data storage. The explanation of the parameters used for this model is as follows:

-Metric: the method used to calculate similarity between data points.
-N-neighbors: the number of neighbors used to classify a data point. Small values tend to overfit, while larger values can cause the model to underfit.
-Weights: defines how neighbors influence the prediction.

The Decision Tree model analyzes past data and combines features to represent possible outcomes based on the dataset. In this case, for example, it will analyze combinations of features such as average monthly income or overtime to determine whether attrition occurred. The advantages of this model include easier interpretation of the decision logic and identification of the most important attributes. However, complex decision trees tend to overfit the data. The explanation of the parameters used for this model is as follows:

–Max_depth: maximum number of levels in the tree.

-Min_samples_leaf: the minimum number of data samples that must be present in a leaf node (a final node in the tree).

-Min_samples_split: the minimum number of data samples required for an internal node to be split into child nodes.

Random Forest is an ensemble method that combines multiple randomly generated decision trees to reach a final prediction based on majority voting. The advantages of this model are that it is a powerful method and reduces the risk of overfitting. However, the training process may take longer compared to other models. The explanation of the parameters used for this model is as follows:

-Max_depth: maximum number of levels in each tree

-Max_features: maximum number of features considered for splitting a node

-Min_samples_leaf: the minimum number of data samples that must be present in a leaf node (a final node in the tree)

-Min_samples_split: the minimum number of data samples required for an internal node to be split into child nodes.

-N_estimators: number of trees

The SVM model constructs a line (or hyperplane) that maximizes the distance between data points from different categories in order to perform classification. The advantages of this model include strong predictive performance and reduced sensitivity to outliers. However, it may show poor performance on very large datasets and requires careful tuning of the kernel and hyperparameters. The explanation of the parameters used for this model is as follows:

-C: refers to the regularization strength used to prevent overfitting.

-Gamma: defines the reach of nearby points affecting the boundary.

-Kernel: defines the hyperplane type

3.3 Initial Models Implemented

The dataset was split into training and test sets using a 67/33 ratio. The split was performed in a stratified manner to preserve the original class distribution.

After the split, we did preprocessing using the One Hot Encoder, in order to encode categorical variables, and did feature scaling. The scaled features were only used in the Logistic Regression, K Nearest Neighbours and Support Vector Machine.

We used five traditional algorithms, and got the following results:

Model	**Configurations**	**Performance**
Logistic Regression	class_weight = balanced max_iter = 3000	AP = 0.56
K-Nearest Neighbours	n_neighbors=10 p = 1 (Manhattan distance)	AP = 0.36
Support Vector Machine	kernel = rbf C = 1.0 custom class weights gamma="scale" probability=True	AP = 0.50
Decision Tree	max_depth=5, min_samples_split=20 min_samples_leaf=10, ccp_alpha=0.001, random_state=42 custom class weights	AP = 0.33
Random Forest	n_estimators=500 max_depth=8 min_samples_leaf=5 max_features="sqrt" random_state=42 n_jobs=-1	AP= 0.47

Also, a grid search was conducted to identify the best parameters and the best metrics for each model. Stratified cross-validation with five folds (Stratified K-Fold, k = 5) was employed. The following table shows the results obtained:

Model	**Parameters**	**Performance**
Logistic Regression	C: 0.1 penalty: l2 solver: liblinear	AP = 0.6174
K-Nearest Neighbours	metric: manhattan n_neighbors: 21	AP = 0.5268

	weights: distance	
Support Vector Machine	C: 0.1 gamma: scale kernel: linear	AP = 0.5965
Decision Tree	max_depth: 3 min_samples_leaf: 1 min_samples_split: 2	AP = 0.3563
Random Forest	max_depth: 5 max_features: sqrt min_samples_leaf: 4 min_samples_split: 2 n_estimators: 100	AP = 0.5722

The first attempt at building the predictive model demonstrated some initial insights but was constrained by several limitations that affected overall performance:

- Limited feature engineering
- Simple class weighting for imbalance data handling
- No advanced sampling techniques
- Suboptimal hyperparameters

3.4 Feature Engineering

In order to tackle the first limitation, new features were created across six categories to capture complex patterns:

Interaction Features, which aims to capture relationships between variables:

New Column	How it was made	Meaning
age_experience_ratio	age / (totalworkingyears + 1)	relationship between age and total experience
income_per_joblevel	monthlyincome / (joblevel + 1)	relationship between income and job lev
overtime_wlb_conflict	overtime * (5 - worklifebalance)	conflict between overtime and worklife balance
commute_burden	distancefromhome * businesstravel_map	impact of commute and business travel
overall_job_satisfaction	jobsatisfaction * environmentsatisfaction	satisfaction in the workplace

New Column	How it was made	Meaning
age_experience_ratio	age / (totalworkingyears + 1)	relationship between age and total experience
income_per_joblevel	monthlyincome / (joblevel + 1)	relationship between income and job lev
career_progression_rate	joblevel / (yearsatcompany + 1)	rate of career progression

Binned Features, creates categorical bins for continuous variables:

New Column	Binns	Column it was made from
age_group	Young (<30), Mid-Career (30-45), Senior (>45)	age
income_quartile	Q1 -Q4	monthlyincome
tenure_group	New, Mid, Experienced, Veteran	yearsatcompany
distance_category	Near, Medium, Far	distancefromhome

Ratio Features, creates ratio features that normalize by relevant denominators:

New Column	How it was made	Meaning
income_per_exp_year	monthlyincome / (totalworkingyears + 1	income per year of experience
promotion_frequency	1 / (yearssincelastpromotion + 1)	frequency of promotions
training_intensity	trainingtimeslastyear / (yearsatcompany + 1)	training intensity
role_stability	yearsincurrentrole / (yearsatcompany + 1)	stability of current role
manager_stability	yearswithcurrmanager / (yearsatcompany + 1)	manager stability
hike_per_level	percentsalaryhike / (joblevel + 1)	percentual salary hike per job level

Satisfaction Features, creates composite satisfaction features:

New Column	How it was made	Meaning
avg_satisfaction	Average of environmentsatisfaction jobsatisfaction, relationshipsatisfaction, worklifebalance	overall satisfaction
satisfaction_variance	Variance of the 4 columns above	consistency of satisfaction
min_satisfaction	Minimum value out of the 4 column	weakest link of satisfaction

low_satisfaction_count	Count of scores <=2 on the 4 columns	consistent dissatisfaction across al areas

Tenure Features, creates advanced tenure-related features:

New Column	How it was made	Meaning
promotion gap ratio	yearssincelastpromotion / (yearsatcompany + 1)	time since last promotion relative to time in company
role stagnation	yearsincurrentrole - yearssincelastpromotion	job role stagnation
early_career	(totalworkingyears <= 5) & (yearsatcompany <= 2)	Binary indicator for new professionals
job_hopper	(numcompaniesworked >= 4) & (yearsatcompany <= 3)	Indicator for frequent job changes

Polynomial Features, creates polynomial features for key numeric variables:

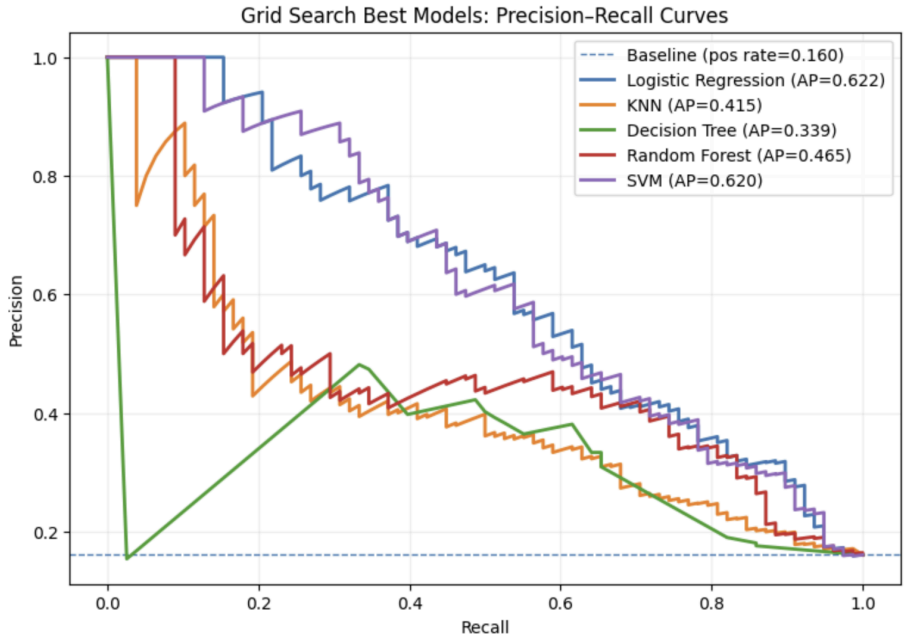
New Column	How it was made
age_squared	square of the age column
totalworkingyears_squared	square of the totalworkingyears column
monthlyincome_squared	square of the monthlyincome column
yearsatcompany_squared	square of the yearsatcompany column

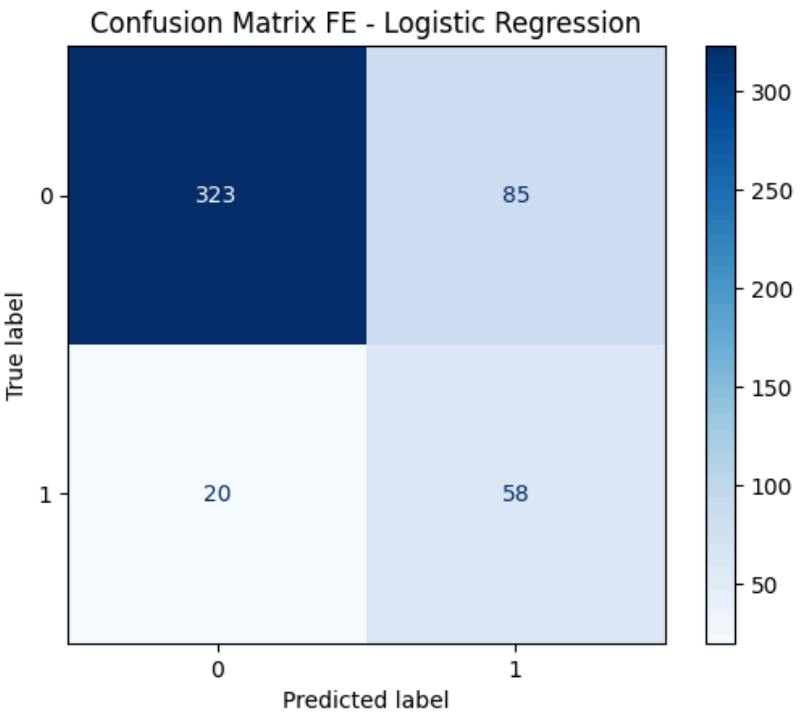
It's important to validate the new columns that were created. In order to ensure there are no redundant features, a Variance Inflation Factor (VIF) analysis was conducted to detect multicollinearity among them. A correlation analysis was also run to identify the features most correlated with attrition. We also ensured the new features were meaningful from a business HR perspective.

Finally, in order to choose the best possible features, we tested different combinations of the aforementioned featuresets on a Logistic Regression model and ranked them using Average Precision. As seen in the table below, the best feature combination includes the given features and polynomial features.

Feature set	Average Precision
baseline + polynomial	0.624867
baseline + polynomial + tenure	0.620840
baseline + tenure	0.615984
baseline	0.608443
baseline + ratio	0.608443
baseline + all_engineered	0.607813
baseline + satisfaction	0.605318
baseline + interaction	0.601934

A Grid Search was again realized using the feature engineering. The precision-recall curves and the confusion matrix for Logistic Regression obtained are the following:





3.5 Imbalanced Data Handling techniques

Several techniques to address data imbalance were tested using a Logistic Regression model, and evaluated using our chosen metric, Average Precision Score, as well as Precision-Recall curves.

Firstly the class weights approach was tested. Instead of changing the data itself, class weighing works by modifying the model's loss function during training, assigning higher importance to minority classes to prevent the model from ignoring them. We also made an attempt where we included the weights in the search grid.

Secondly, multiple Undersampling techniques were tested. Undersampling works by reducing the majority class in order to make the dataset more balanced. We tried Random Undersampling, Tomek Links, Edited Nearest Neighbours (ENN) and Near Miss.

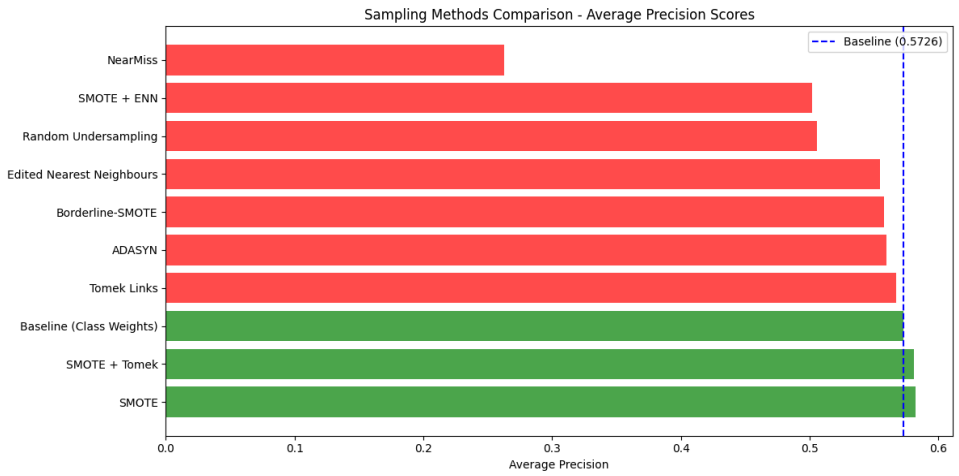
Oversampling techniques were also tested, using multiple SMOTE variants. We tried Standard SMOTE, Borderline SMOTE and ADASYN.

Finally, we tried combined approaches, namely SMOTE with Tomek Links and SMOTE with ENN.

The following table shows the best method per model:

	Method	Model	Average Precision	Class Dist (Train)	Improvement vs No Sampling
0	ENN	Decision Tree	0.384860	{0: 477, 1: 159}	0.031235
1	ENN	KNN	0.395716	{0: 625, 1: 159}	0.020392
2	SMOTE	Logistic Regression	0.581985	{0: 825, 1: 825}	0.009387
3	ADASYN	Random Forest	0.563493	{0: 825, 1: 834}	0.077267
4	Tomek Links	SVM (RBF)	0.542122	{0: 792, 1: 159}	0.007810

The following graph shows the best methods for Logistic Regression, according to Average Precision Score.



4. Results and Discussions

From all the tests performed and considering precision–recall as the chosen metric, the best result was achieved by the logistic regression model with feature engineering, reaching an AP = 0,622. The improvement compared to the baseline suggests that this combination was able to capture relevant patterns related to attrition, and that the feature engineering introduced relationships that were not clearly visible in the original dataset.

Top Predictive Features (Expected)

During feature engineering, it was also possible to rank the top features expected to be most relevant for predicting attrition. The features are listed below:

- Overtime + Work-Life Balance interactions
- Monthly Income relative to job level
- Years since last promotion
- Job satisfaction metrics
- Age-experience alignment
- Stock option level
- Distance from home + travel frequency
- Number of companies worked
- Years with current manager
- Training times last year

Some high-risk indicators were also identified during feature engineering, which may signal potential attrition. The table below presents these indicators along with suggested actions to help prevent employees from leaving.

High Risk Indicator	Suggestion
Overtime without work-life balance	Review workload
Low career progression	Accelerate promotions
Compensation misalignment	Salary benchmarking
Low satisfaction scores	Engagement initiatives
Long commute + frequent travel	Remote work options

Intervention Strategy

By considering different probability thresholds, it becomes possible to design strategies to manage and prevent employee attrition. In this analysis, a probability higher than 0.7 is interpreted as a high risk of an employee leaving the company, while a probability below 0.4 represents a low-risk scenario that requires only mild interventions. The table below summarizes these threshold values and the corresponding recommended actions to mitigate attrition.

Risk	Probability of Attrition	Strategies
High	$P > 0,7$	-Immediate manager intervention -Retention bonus consideration -Weekly check-ins
Medium	$0,4 \leq P \leq 0,7$	-Monthly discussions -Training opportunities -Quarterly reviews
Low	$P < 0,4$	-Standard engagement programs -Annual reviews

ROI Estimation

Considering that, after feature engineering, 16% of employees were classified as having left the company, applying this percentage to the original dataset would correspond to 235 employees at risk of attrition. With a recall of 74%, the model would correctly identify 174 of these cases. Then, given a successful intervention of 40%, the model would ultimately succeed in 69 true cases of attrition.

Assuming that each retained employee represents a cost saving of $50,000, this would result in an estimated annual savings of $3.5 million for the company.

Employees at risk: 235 (16% of 1,470)
Identified by model (74% recall): 174
Successful interventions (40% success): 69
Cost savings: 69 × $50,000 = $3.5M annually

Limitations and Future Work

Although the insights obtained are very interesting and meaningful, it is important to acknowledge that the project has limitations due to the relatively small dataset. This occurs because the data is based on sensitive company information, making it difficult to access real

and complete datasets. In addition, employee attrition is not always tracked consistently over long periods, which limits the availability of extended historical data.

The dataset also lacks time-series features, and correlations with external market conditions are not captured. As a result, although economic factors have a significant impact on attrition, they are not considered in the current analysis. Furthermore, the satisfaction survey may contain bias, as employees might not express their true feelings in an attempt to please researchers or the company.

For future work, it would be valuable to use a larger dataset with more comprehensive information, including features such as the month and year in which each employee left the company, as well as external indicators like inflation or unemployment rates. These additions would help capture both temporal dynamics and broader economic influences.

5. References & Resources:

Enrich. (2024). The Cost of Replacing an Employee and the Role of Financial Wellness. Available in: https: //enrich.org/the-true-cost-of-employee-turnover-financial-wellness-enrich/

Fang, Y., & Zhang, Z. (2025). Employee Turnover Prediction Model Based on Feature Selection and Imbalanced Data Handling. IEEE Access, 20. https://doi.org/10.1109/ACCESS.2025.3589491

Amzad Basha, M. S., Varikunta, O., Devi, A. U., & Raja, S. (2024). Machine Learning in HR Analytics: A Comparative Study on the Predictive Accuracy of Attrition Models. 2024 2nd International Conference on Device Intelligence, Computing and Communication Technologies (DICCT), 475–480. https://doi.org/10.1109/DICCT61038.2024.10533064

Scikit-learn developers. (n.d.). *sklearn.utils.class_weight.compute_class_weight*. Scikit-learn. https://scikit-learn.org/stable/modules/generated/sklearn.utils.class_weight.compute_class_weight.html

Imbalanced-learn Developers. (2024). Under-sampling [Version 0.14.1]. Imbalanced-learn. https://imbalanced-learn.org/stable/under_sampling.html

Shakir, Omar. (2025). Fairness-Aware Mutual Information for Feature Selection in Employee Attrition Prediction. Wasit Journal of Computer and Mathematics Science, 4(4), 26–37. https://doi.org/10.31185/wjcms.404

Naik, P., & Chakraborti, S. (2025). Employment tenure prediction for Indian IT professionals using linkedin Network Data and machine learning based techniques. Discover Analytics, 3(1). https://doi.org/10.1007/s44257-025-00052-3

P, J. P., D, V., & K. V, U. (2022). Effective classification of IBM HR analytics employee attrition using sampling techniques. 2022 Second International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT), 1–6. https://doi.org/10.1109/icaect54875.2022.9808057